

Geography — "Our Restless Earth" Key Notes

無止境的地球重點筆記 · Unit Test — **Thu 15 Oct 2026**, 40 minutes, in class · Geo 6B (Mr Benny Tse)
Revision scope: workbook pp. 3–17 · PPTs 0 / 1 / 2a / 2b / 3 · Levels of Questioning sheet · map skills
(grid refs, compass)

 **Source note 來源說明** : these notes are built from the **actual teaching PPTs** (Lessons 0, 1, 2a, 2b, 3) read in Google Classroom — definitions quoted are the teacher's own words. The **physical lesson workbook (pp. 3–17) is NOT posted online**, so workbook-only reading articles are not covered here.
本筆記根據課堂 PPT (Lesson 0/1/2a/2b/3) 編寫，定義為老師原話。實體練習簿 (第 3–17 頁) 未上載到網上，故簿內補充閱讀文章不在此列。

1. Key Terms 關鍵詞彙

Plate 板塊 — "The earth's surface land mass is divided up into pieces called **plates**."
地球表面陸地被分割成稱為「板塊」的碎片。

Plate boundary 板塊邊界 — "The **narrow junction** between two moving plates is called a plate boundary."
兩個移動板塊之間狹窄的交界稱為板塊邊界。

Lithosphere 岩石圈 — "Consists of the **crust and the uppermost part of the mantle; rigid; solid**." Recall Q&A: (a) the two parts are the **crust** and the **uppermost part of the mantle**; (b) they **float and move slowly on the semi-fluid asthenosphere / mantle**. The **continental crust is thicker**; the **oceanic crust is under the sea and thinner**. The crust is "**made up of rigid rocks**".
岩石圈 = 地殼 + 地幔最上層；堅硬；固體。浮在半流動的軟流圈/地幔上緩慢移動。大陸地殼較厚，海洋地殼較薄。

Convection currents 對流 — the driving force: **hot material rises, cold material sinks** (the teacher's hot-air-rises / cold-air-sinks analogy). "The **convection cell drives seafloor spreading and moves the tectonic plates**."
對流 = 熱物質上升、冷物質下沉；對流推動海底擴張並移動板塊。

Subduction 隱沒 — in the Himalayas story: "**Oceanic crust subducting under Asian continent**" — one plate slides beneath another. "Over time, **all the oceanic crust of the Indian subcontinent subducted**."
隱沒 = 一個板塊滑到另一個板塊之下（如大洋地殼隱沒到亞洲大陸之下）。

2. Three Plate Movements 三種板塊運動

Movement 板塊運動	Internal force 內力	Boundary 邊界	Arrow 方向箭咀
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Move apart 互相遠離	Tension 張力	Divergent 分散性	←→ plates move away from each other
Move towards 互相靠近	Compression 擠壓力	Convergent 聚合性	→← plates move into each other
Move past 互相擦過	Shear 剪切力	Transform 保守性	↑ ↓ plates slide past each other

- The teacher's force wording: **Compressional force** — "Generated at **convergent** plate boundary. The rocks become **shorter and thicker**." 擠壓力令岩石變短變厚。
- **Tensional force** — "Generated at **divergent** plate boundary. The rocks become **longer and thinner**." 張力令岩石變長變薄。
- **Shearing force** — "Generated at **Conservative (transform)** plate boundary. The rocks **twist in shape**." 剪切力令岩石扭曲。 ⚠ Only the transform boundary gets the "conservative" alternate name — **constructive / destructive were NOT used** by the teacher.

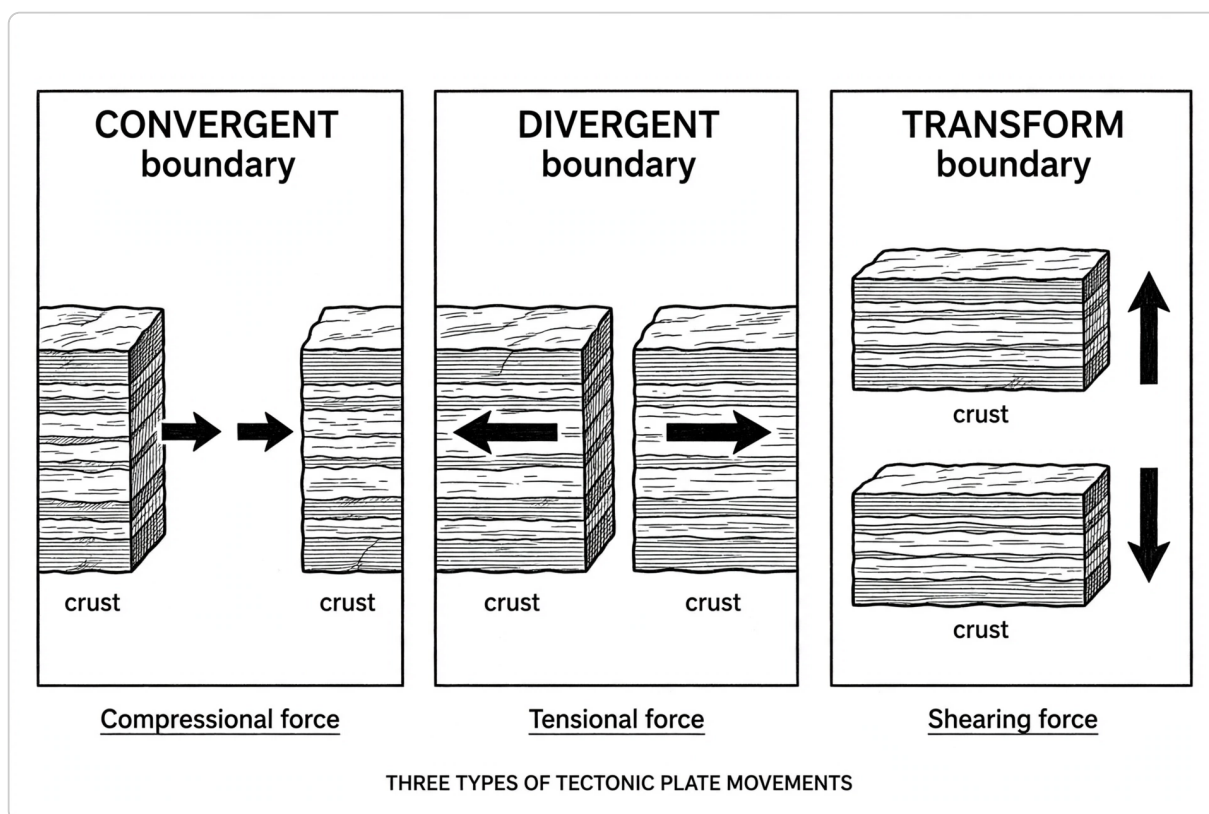


Fig. 1 — The three plate movements, forces and boundaries (match arrows: →← convergent, ←→ divergent, ↑ ↓ transform).

3. Internal Processes 內部過程

"The internal forces mentioned before give rise to three internal processes, namely **folding, faulting and volcanism**."

內力產生三種內部過程：褶曲、斷層、火山活動。

Folding 褶曲 — "Rock layers are **folded up under compressional force**. This process is folding."

岩層在擠壓力下向上彎曲成褶皺。

Faulting 斷層 — "Faulting refers to the process whereby **rocks are broken and displaced relative to each other**. The **cracks or fractures along which rocks move are called faults**."

岩石斷裂並相對移位；岩石移動的裂縫稱為斷層線。

Volcanism 火山活動 — "Volcanism refers to **all the processes related to the occurrence and the movement of magma**."

所有與岩漿出現及移動有關的過程。

4. Fold Mountains — the Himalayas 褶曲山脈：喜馬拉雅山

★ **Test favourite:** the teacher tells the Himalayan story in 3 stages — learn them in order.

- **Stage 1 — ocean-continent collision 海洋-大陸碰撞**：the Indian subcontinent and the Asian continent were separated by a sea. **Sediments were deposited at the land margin and compacted into sedimentary rock**. The **oceanic crust subducted under the Asian continent**, and **volcanoes were formed as magma rose through cracks**.
- **Stage 2 — continent-continent collision 大陸-大陸碰撞**："Over time, **all the oceanic crust of the Indian subcontinent subducted**" — direct collision of the two continents.
- **Stage 3 — folding 擠壓成山**：the rocks at the plate boundary were **crumpled into the Himalayas**. "Under compressional force, **folding and faulting occur**."

FOLD MOUNTAIN FORMATION

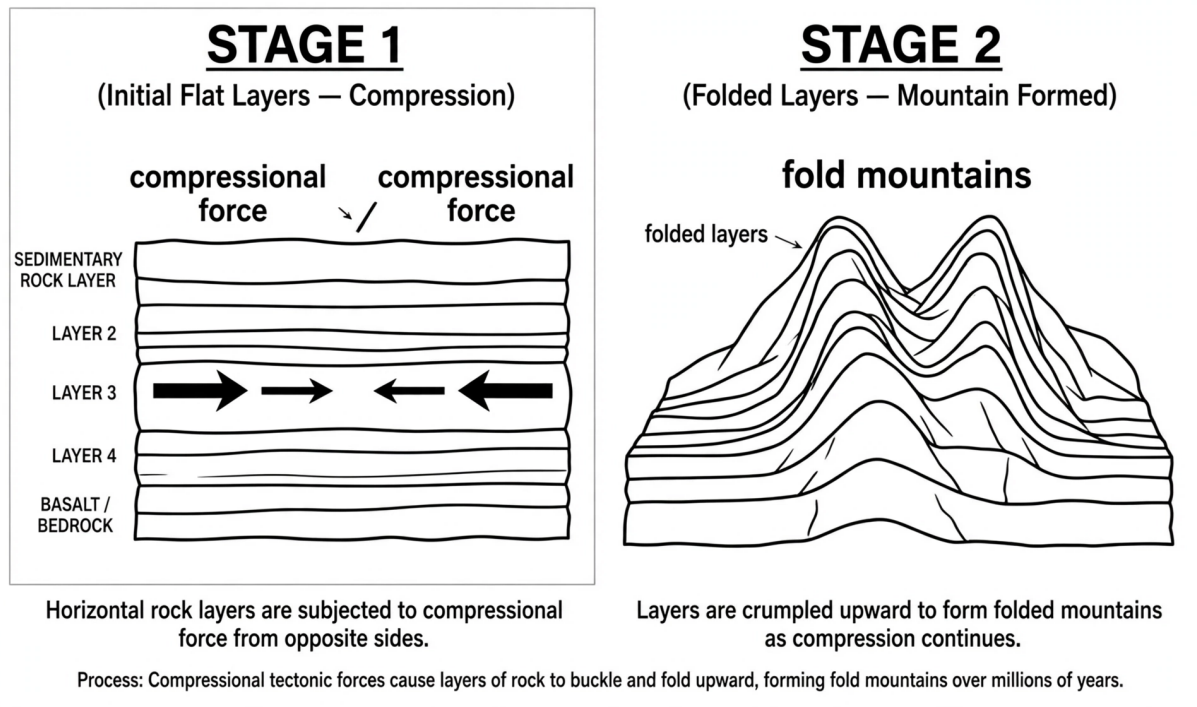


Fig. 2 — Fold mountain formation: horizontal layers → squeezed by compressional force → crumpled upward into fold mountains.

- **Highest peak 最高峰** : "Qomolangma Feng (8,848 m), the highest peak of the Himalayas."
- **Other fold mountains 其他褶曲山脉** : the **Atlas Mountains** (Toubkal, 4,165 m), the **Rockies**, the **Andes** (Mount Aconcagua, 6,960 m).

5. Volcanoes 火山

Crater 火山口 — "At the top of a volcano, there is a **circular depression called a crater.**"
火山頂部的圓形凹陷稱為火山口。

Vent 火山口通道 = the channel through which magma rises to the crater.

Magma chamber 岩漿庫 = the underground reservoir where magma collects before rising.

Vent = 岩漿上升的通道；Magma chamber = 岩漿聚集的地下儲庫。

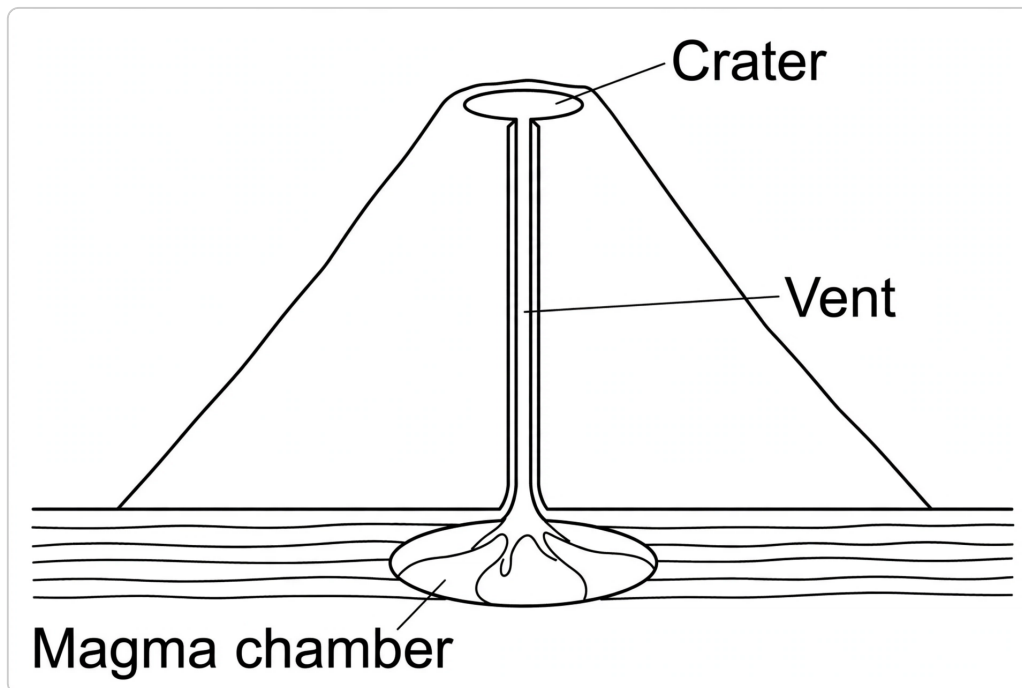


Fig. 3 — Volcano cross-section: Crater → Vent → Magma chamber.

- **Intrusive volcanism 侵入性火山活動**：magma that cools underground. "**Batholith — a dome-shaped chamber, formed deep underground.**" 岩基 = 深埋地下的穹形岩體。
- **Distribution 分佈**：active volcanoes are found **on plate boundaries**, including the **Pacific Ring of Fire** — the ring of volcanoes around the Pacific Ocean. **Data-response sentence**: "Active volcanoes are mainly distributed along plate boundaries, especially around the rim of the Pacific Ocean (the Pacific Ring of Fire)."

6. Plates on the Teacher's Map 板塊圖

- The 14 plates on the teacher's map: **Eurasian, Arabian, African, Indo-Australian, Antarctic, Okhotsk, Juan de Fuca, North American, Caribbean, Philippine, Pacific, Cocos, South American, Nazca.**
- ★ **Hong Kong is on the Eurasian Plate, near the Philippine Plate boundary.** 香港位於歐亞板塊，接近菲律賓板塊邊界。

7. Map Skills 地圖技巧

Compass 羅盤 : 16-point compass (N, NNE, NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, NNW). The teacher's method: **Step 1 — draw a straight line between the two places**, then set a compass rose at the starting place and read the direction.

十六方位羅盤：先畫一條連接兩地的直線，再在起點放羅盤讀方向。

Grid references 網格座標 : "Grid references are **grid systems for locating places or features** on maps and plans."

Eastings 東距 = the **vertical** grid lines; numbers **increase eastward**.

Northings 北距 = the **horizontal** grid lines; numbers **increase northward**.

東距 = 垂直線，向東數字增大；北距 = 水平線，向北數字增大。

- **Four-figure grid reference 四位網格座標** : read **EASTINGS first, then NORTHINGS** ("go along the corridor, then up the stairs" — 先橫後直). Find the square the feature is IN. Teacher's examples: locate the bridge, the temple, the beacon, the golf course, the country park management centre on a HK map extract.
- ⚠ Eastings lines run north-south (vertical) but the number tells you how far **east** you are — don't mix the two up.

8. Levels of Questioning 提問層次

Level	Meaning 意思	Example 例子
L1 = Look	Just describe — what can you see?	L1: What can you see at the top of the volcano?
L2 = Why?	Explain — the reason / cause	L2: Why do fold mountains form at convergent boundaries?
L3 = What if?	Imagine! — predict / imagine a change	L3: What if two plates stopped moving? What would happen to volcanoes?

- Tips: **no vague words** like "it" / "thing" — name the thing; **no yes/no closed questions** — ask open-ended ones; use **geographic words** (ash, magma, plate boundary, eruption, environment); **write "L1 / L2 / L3" after each question** you make.

★ **Test-day checklist 測驗日檢查清單** : ① Plate = pieces of the earth's surface land mass. ② Lithosphere = crust + uppermost mantle; rigid; solid. ③ Movement→force→boundary: apart = tension = divergent; towards = compression = convergent; past = shear = transform (also called conservative). ④ Arrows: →← convergent, ←→ divergent, ↑↓ transform. ⑤ Himalayas 3 stages in order (ocean-continent → continent-continent → crumple/fold). ⑥ Qomolangma Feng = 8,848 m. ⑦ Crater = circular depression at the top. ⑧ Volcanoes sit on plate boundaries — Pacific Ring of Fire. ⑨ HK = Eurasian Plate, near the Philippine Plate boundary. ⑩ Grid refs: eastings first, then northings.